

Challenge (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x + 1$, what is $(f + g)(x)$?
(A) $x^2 + 2x + 1$
(B) $x^2 - 2x - 1$
(C) $x^2 + 2x - 1$
(D) $2x^2 + x + 1$
(E) $x^2 - 2x + 1$
- Q2.** A chef mixes $f(x) = 2x$ cups of flour with $g(x) = x + 1$ cups of sugar to make a cake. What is $(f - g)(x)$?
(A) $x - 1$
(B) $x + 1$
(C) $2x - x - 1$
(D) $2x + x + 1$
(E) $x - 2$
- Q3.** Given $f(x) = x - 2$ and $g(x) = 3x + 1$, what is $(f \cdot g)(x)$?
(A) $3x^2 - 5x - 2$
(B) $3x^2 + 5x - 2$
(C) $3x^2 - 5x - 3$
(D) $3x^2 + 5x + 2$
(E) $x^2 - 5x - 2$
- Q4.** The domain of $f(x) = \frac{1}{x}$ is all real numbers except $x = 0$. If $g(x) = 2x$, what is the domain of $(f \cdot g)(x)$?
(A) All real numbers
(B) All real numbers except $x = 0$
(C) All real numbers except $x = 1$
(D) All real numbers except $x = 2$
(E) All real numbers except $x = -1$
- Q5.** A construction worker needs $f(x) = x + 2$ feet of wood and $g(x) = 2x$ feet of pipe. What is $(f + g)(x)$?
(A) $3x + 2$
(B) $3x - 2$
(C) $x + 2$
(D) $2x + 2$
(E) $x - 2$
- Q6.** Given $f(x) = x^2 - 1$ and $g(x) = x + 2$, what is $(f - g)(x)$?
(A) $x^2 - x - 3$
(B) $x^2 + x + 1$
(C) $x^2 - x + 1$
(D) $x^2 + x - 3$
(E) $x^2 - x - 1$
- Q7.** The area of a kitchen is given by $f(x) = x^2$. If the length of the counter is $g(x) = 2x$, what is $(f \cdot g)(x)$?
(A) $2x^3$
(B) $2x^2$
(C) x^3
(D) x^2
(E) $2x$
- Q8.** A recipe calls for $f(x) = x$ cups of flour and $g(x) = x - 1$ cups of sugar. What is the domain of $(f + g)(x)$ if x must be a positive integer?
(A) $x \geq 1$
(B) $x \geq 0$
(C) $x > 0$
(D) $x \geq 2$
(E) $x > 1$

Open Ended Questions (FRQ)

- Q9.** If $f(x) = x^2 + 1$ and $g(x) = 2x - 1$, find $(f \cdot g)(x)$ and simplify the resulting expression.
- Q10.** A builder needs to design a house with a floor area given by $f(x) = x^2$ and a roof area given by $g(x) = 2x$. Find the domain of $(f + g)(x)$ if x must be a positive real number and $x \leq 10$.

Chef's Tips

- Remind students to show all work and simplify expressions.
- Emphasize the importance of checking the domain of combined functions.
- Encourage students to use real-world examples to illustrate their understanding of function operations.